

Loops and Patterns in Java

20 Coding Practice Questions

For Loop

1. Print all numbers from 10 down to 1 using a for loop. *(Beginner)*
2. Print all odd numbers between 1 and 20. *(Beginner)*
3. Find the factorial of a number entered by the user (e.g., $5! = 120$). *(Easy)*
4. Print the squares of numbers from 1 to 10 (1, 4, 9, 16, ...). *(Easy)*
5. Print the multiplication tables of 2 to 5, one after another, each clearly labeled. *(Medium)*

While Loop

6. Print numbers from 1 to n (n taken as input) using a while loop. *(Beginner)*
7. Count how many digits a given number has (e.g., 4527 has 4 digits). *(Easy)*
8. Reverse a given number (e.g., 1234 becomes 4321) using a while loop. *(Easy)*
9. Keep asking the user to enter a number until they enter 0, then print the sum of all numbers entered. *(Medium)*
10. Find the digital root of a number by repeatedly summing its digits until a single digit remains (e.g., 3875 to 23 to 5). *(Medium)*

Nested Loops

11. Print a 4x4 grid of numbers where every row contains 1, 2, 3, 4. *(Easy)*
12. Print the full multiplication table from 1 to 10 using nested loops (a 10x10 grid). *(Medium)*
13. Given a 3x3 matrix (2D array) of integers, print it row by row using nested loops. *(Medium)*
14. Using nested loops, print all pairs (i, j) where $1 \leq i, j \leq 5$ and $i + j$ is exactly 6. *(Hard)*

Pattern Programming

15. Print a hollow square pattern of size n (stars only on the border, spaces inside). *(Easy)*
16. Print a diamond pattern by combining a pyramid with an inverted pyramid below it. *(Easy)*
17. Print a number pattern where each row counts down instead of up (e.g., 4 3 2 1 / 3 2 1 / 2 1 / 1). *(Medium)*
18. Print a reverse alphabet triangle (e.g., D C B A / C B A / B A / A). *(Medium)*
19. Print a Pascal's Triangle-style number pattern up to n rows, where each number is the sum of the two numbers above it. *(Hard)*

Mixed Challenge

20. Write a program that prints a hollow pyramid (stars only on the outer edges and base, spaces inside). (*Challenge*)